

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} Q(x) &= \int_{x/\log x}^{x-\sqrt{x}} \frac{1}{\log(x-t)} \left(\frac{t}{\log t} + O\left(\frac{t}{\log^2 t}\right) \right) dt + O\left(\frac{x^2}{\log^3 x}\right) \\ &= \int_{x/\log x}^{x-\sqrt{x}} \frac{t}{(\log t) \log(x-t)} dt + O\left(\int_{x/\log x}^{x-\sqrt{x}} \frac{t}{(\log^2 t) \log(x-t)} dt + \frac{x^2}{\log^3 x}\right). \end{aligned}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} (f_S)_\lambda(s_1, s_2) &\leq \sup_{(y'_1, y'_2, x') \in \mathcal{S}} \left\{ f_1(y'_1) + f_2(y'_2) - \sum_{1 \leq \ell \leq 2} \lambda_\ell c_{Y_\ell}(y_\ell, y'_\ell) \right\} \\ &= (f_1)_{\lambda_1}(y_1) + (f_2)_{\lambda_2}(y_2). \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} K_{\omega, c}(\lambda \Phi_\omega) &= 2\lambda^2 L(\Phi_\omega) + 3\lambda^3 N(\Phi_\omega) + 2\lambda^2 \omega Q(\Phi_\omega) + 2\lambda^2 cP(\Phi_\omega) \\ &= \lambda^2 K_{\omega, 0}(\Phi_\omega) + \lambda^2 \{3(\lambda - 1)N(\Phi_\omega) + 2cP(\Phi_\omega)\} \\ &= 2\lambda^2 \{-3(\lambda - 1)\mu_{\omega, 0} + cP(\Phi_\omega)\}. \end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} f^{B, \alpha}(\xi) &:= \partial_x^\alpha|_{x=0} \alpha_{-x}(f(\xi + Bx)) \\ &= \sum_{\beta + \gamma = \alpha} \binom{\alpha}{\beta} i^{|\beta|} \delta^\beta (\partial_x^\gamma|_{x=0} f(\xi + Bx)) \\ &= \sum_{\beta + \gamma = \alpha} \binom{\alpha}{\beta} i^{|\beta|} \delta^\beta \partial_{B, \xi}^\gamma f(\xi). \end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \epsilon &= 1 & \delta \varepsilon_1^1\left(\frac{\pi}{5}\right) &= \delta \varepsilon_3^1\left(\frac{\pi}{5}\right), \\ & & \delta \varepsilon_1^1\left(\frac{\pi}{4}\right) &= \delta \varepsilon_2^1\left(\frac{\pi}{4}\right), \\ & & \delta \varepsilon_0^1\left(\frac{\pi}{4}\right) &= \delta \varepsilon_3^1\left(\frac{\pi}{4}\right), \end{aligned}$$